Md Omar Faruqe Riyad

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Education

Shahjalal University of Science and Technology (SUST) — Sylhet, Bangladesh
B.Sc. in Computer Science and Engineering; CGPA: 3.53/4.00 (Last two years: 3.83) — *2018 – 2023*

Experience

Dynamic Solution Innovators — April 2023 – Present
Software Engineer — *Dhaka, Bangladesh*

- Engineered security-critical government applications for NBR, MoCAT, and BCIC, transforming complex offline bureaucracies into automated digital workflows using **Java EE (JSF, EJB, JPA)**.
- Open-sourced `gf`, a cross-platform CLI dev tool with **JDWP bytecode hot-swap** (via JDI API), incremental compilation, and JasperReports classloader cache bypass, reducing edit-test cycles by **96% (2m → 5s)**. Integrated with **Coding Agent** through custom skills for autonomous server management.
- Designed an event-driven architecture using **JMS** to decouple services, ensuring high availability and fault tolerance for the E-Appeal system.
- Engineered a centralized monitoring stack (**Prometheus, Loki, Grafana**) and dockerized the full application stack for consistent local development.
- Contributed to the design of a **state-driven approval workflow** for the Smart Hotel Management System (SHMS), reusable across multiple application types.
- Led end-to-end delivery of the **Procurement, Inventory, and Asset** module for BCIC-ERP, from business analysis (As-Is/To-Be modeling) and database schema design to production deployment.
- Conducted vulnerability assessments and security audits as a key member of the internal cybersecurity team.

Altri.ai — Part-time
Full-Stack AI Engineer — *Remote, US*

- Architected a hybrid AI scoring engine combining **LLM-based feature extraction** from unstructured property descriptions with traditional ML models for price prediction and investment recommendations.
- Refactored data enrichment pipeline using **Python asyncio**, achieving a **10x throughput increase** and reducing cloud compute costs by parallelizing I/O-bound LLM operations.
- Built a fault-tolerant data ingestion engine using HomeHarvest and **PostgreSQL**, implementing a "Delta Sync" strategy with immutable event logging to track property lifecycles without data loss.
- Established a containerized training pipeline with Human-in-the-Loop validation gates, ensuring new "Challenger" models outperform production "Champions" before deployment.
- Developed a FastAPI backend and React frontend backed by Supabase. Implemented robust security via Row Level Security (RLS) and distributed job locking for safe, multi-tenant concurrency.
- Dockerized the complete microservices stack (API, Scheduler, Trainer) for reproducible deployments across local environments and Render cloud infrastructure.

Projects

KnowledgeRelay | *Python, FastAPI, LangChain, ChromaDB, React* — 2025

- Built a RAG system to preserve project-specific knowledge bases and assist team members via contextual Q&A.
- Engineered a multi-stage knowledge ingestion pipeline that automatically generates QA pairs from documents.
- Integrated **ChromaDB** with vector embeddings and metadata-based filtering for efficient retrieval.
- Implemented contextual query rephrasing using chat history for higher accuracy.

PrimeFaces Open Source Contribution | *Java, JSF, PrimeFaces* — 2024

- Contributed a feature enhancement to the `JPALazyDataModel` of **PrimeFaces** component library.
- Enabled developers to perform complex, customizable filtering within DataTables.

Shopaholic | *JavaScript, Node.js, Express.js, MongoDB, React, Redux* — 2022

- Built a full-stack e-commerce platform connecting customers, banks, and suppliers.

Brevity | *Python, Flask, MySQL, JavaScript* — 2021

- Developed a knowledge-sharing platform connecting beginners with experts in various fields.

Research & Publications

AdversaryAI at BLP-2025 Task 2 BLP 2025, IJCNLP-AACL

- **Low-Resource Adaptation:** Investigated low-resource language code generation by fine-tuning open-source models (Llama 3, TigerLLM, Qwen) using parameter-efficient **LoRA** adapters.
- **Reasoning Enhancement:** Enhanced model reasoning via **Chain-of-Thought (CoT)** and a self-refinement loop that iteratively critiques and corrects generated code based on execution feedback.
- **Error Analysis:** Conducted rigorous error analysis to identify failure modes, achieving **85% accuracy** and **4th Place** out of 32 teams.

Deja Vu at SemEval 2024 Task 9 SemEval 2024, NAACL

- Designed data augmentation pipelines to improve model robustness for common sense reasoning tasks.
- Conducted a comparative study of Advanced Language Models, analyzing performance gaps in lateral thinking puzzles.

Team_Syrax at BLP-2023 Task 1 BLP 2023, EMNLP

- **Semi-Supervised Learning:** Applied self-training techniques and back-translation to mitigate data scarcity and class imbalance in Bangla text.
- **Ensemble Strategy:** Implemented bagging and majority voting ensembles for Violence Inciting Text Detection, achieving **Top 20** rank.

Visual Question Answering (VQA) Data Synthesis | *Undergraduate Thesis* 2022

- Addressed "Answer Space Diversity" limitation in VQA v2 by augmenting training data with automatically generated, contextually relevant QA pairs.
- Conducted a comprehensive study of VQA methods to analyze limitations in visual reasoning.

Honors, Awards & Certifications

4th Place out of 32 teams – Code Generation in Bangla Shared Task, IJCNLP-AACL 2025 [Leaderboard]

6th Place out of 30 teams – DSI AI Agent Hackathon, 2025 [Leaderboard]

Top 20 (Rank 12 post-eval) – Violence Inciting Text Detection (VITD), BLP 2023 [Leaderboard]

Ranked 11th – BUET CSE Fest 2022 AI Contest

Certified in Cybersecurity (CC) – ISC2 [Badge]

Solved 500+ algorithmic problems on Codeforces, LeetCode, and LightOJ.

Leadership & Teaching

Course Instructor 2024 – Present
Dynamic Learning *Remote*

- Authored and instructed **Enterprise Web Development with JEE, JSF, and PrimeFaces** course, teaching core architectural patterns including component-driven design, CDI/EJB integration, and JSF lifecycle management.

Volunteer 2019

- Served as volunteer at IEEE International Conference on Bangla Speech and Language Processing (ICBSLP).

Technical Skills

Languages: Java, Python, C/C++, SQL, JavaScript, Shell
Backend & Systems: FastAPI, Flask, Java EE (JSF, JPA, CDI, EJB), GlassFish, JMS
Frontend: React, PrimeFaces
AI & Data Engineering: PyTorch, LangChain, LLM, RAG, ChromaDB
Database & Cloud: PostgreSQL, MySQL, Neo4j, Supabase, Docker, GitHub Actions, MinIO
Tools & Observability: Prometheus, Grafana, Loki, Git, Linux, ZFS, Flyway, JasperReports, LaTeX